Project

Development of a Hybrid Classification Model Using Dynamic Hybrid Distance Metrics and a KNN-Random Forest Ensemble

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Problem Definition

The performance of machine learning models depends heavily on both algorithm design and the compatibility between model choices and data characteristics. The K-Nearest Neighbors (KNN) algorithm is notable for its simplicity and ability to capture local patterns. However, its performance is highly sensitive to the choice of distance metric. On the other hand, Random Forest (RF) is a powerful ensemble method for capturing global patterns, yet it may lack sensitivity to local data structures.

Current ensemble approaches often overlook the dynamic integration of models with fundamentally different characteristics—such as lazy learning-based KNN and tree-based RF. Furthermore, using a fixed distance metric in KNN limits its generalization capacity across diverse data types.

This project aims to enhance KNN performance by developing dynamic hybrid distance metrics tailored to dataset characteristics. It also proposes a novel hybrid classification model by integrating the improved KNN with Random Forest using a dynamically weighted ensemble approach. The project contributes two key algorithmic innovations:

1. **KNN with Dynamic Distance Metrics**: Automatic selection of distance metrics (Euclidean/Manhattan/Cosine) based on data features.

2. **Hybrid Ensemble with Hyperparameter Integration**: A weighted ensemble model combining KNN and RF, with dynamic weight adjustment based on performance.

Approach And Algorithms

1. **Algorithmic Contributions**
   The project is built upon two core algorithmic innovations:

a) KNN with Dynamic Distance Metric

- **Metric Selection Mechanism**: A rule-based decision function will be designed by analyzing the statistical properties of the dataset (e.g., dimensionality, feature correlation, class entropy, and density distribution). For example:
  - If dimensionality > 50 and correlation < 0.3 → Use Cosine Similarity
  - If dimensionality ≤ 50 and class entropy > 1.5 → Use Manhattan Distance
  - Otherwise → Use Euclidean Distance
- **Dynamic Weighting**: A confidence score for the selected metric (e.g., based on validation performance) will be used as a weight in the neighbor voting process.

- **Hyperparameter Optimization**: The number of neighbors (k) and thresholds for metric selection will be adaptively optimized using Bayesian Optimization.

b) Hybrid Ensemble with Integrated Hyperparameters SİL. KNN SEÇİLİRSE YANİ.

- **Dynamic Weighting Mechanism**: Local similarity scores from KNN and global decision votes from Random Forest will be dynamically combined based on model uncertainty (e.g., entropy or variance) observed in the validation set.
- **Integration Strategy**: Class probabilities generated by KNN for each test instance will be combined with those from RF using a weighted voting scheme. Weights will be updated according to each model's validation performance.
- **Hyperparameter Optimization**: Parameters such as k and distance metric for KNN, and number of trees and tree depth for RF, will be optimized using Grid Search and Randomized Search.

The project will utilize three publicly available datasets from the UCI Machine Learning Repository:

1. **Wine Quality Dataset**

This dataset is used to predict the quality of wines based on physicochemical properties. It contains 4,898 instances. On the red wine subset, KNN (with k=5 and normalization) achieves an accuracy of approximately 65–70%.On the white wine subset, KNN performs slightly better, with an accuracy around 70%.Random Forest achieves 85–88% accuracy on red wine and 87–90% on white wine.

2. **Mushroom (Agaricus-Lepiota) Dataset**

This dataset classifies mushrooms as edible or poisonous based on 22 categorical attributes (e.g., cap shape, odor, gill color). It includes 8,124 instances. KNN shows strong results, with an accuracy between 90–95%.

3. **Dry Bean Dataset**

This dataset involves classifying 7 different types of dry beans using 16 image-based features (e.g., shape, size, color, texture), extracted from bean images. It includes 13,611 instances.KNN achieves around 91.2% accuracy, while Random Forest yields approximately 96%.

References

- https://archive.ics.uci.edu/datasets
- Mushroom - UCI Machine Learning Repository
- https://archive.ics.uci.edu/ml/datasets/wine+quality
- https://archive.ics.uci.edu/dataset/602/dry+bean+dataset
- https://scikit-learn.org/stable/modules/neighbors.html
- https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

YAPILACAKLAR;

Title DİSTANCE METRİC LEARNİNG BAŞLIĞI BU OLSUN.. PAPER VARMIŞ bununla ilgili ,araştır.

a) Kısmında, **KNN with Dynamic Distance Metrics 2 SİNDEN BİRİNİ (RF ya da KNN)SEÇ, KNN OLANI SEÇİN öneri olarak dedi HOCA, DİSTANCE METRİC LEARNİNG'E BAKın dedi.**

Datasets SİMPLE DATASETS, MORE REALİSTIC bulmalsıın LARGER DATASETS, olacak.

UCI dekiler çok SMALL, simple BAŞKA VERİ SETİ BUL.
FACE RECOGNITİON GİBİ MESELA

UPDATE THE PROPOSAL AND SEND MAİL TO HOCA next week. SADECE HOCAYA MAİL ATIN PROPOSAL GÜNCELLEYİP.

KOD KULLANMADA SORUN YOK, BAŞKALARINI KODLARINI KULLANDIĞIN ZAMAN CİTATİON YAP MUTLAKA. Kodu burdan/şurdan aldım falan ekle
PROJECT TİTLE'ı change falan olunca google sheets'e yaz 1 sutün ekle project titleını oraya yaz.

INDERMİDATE REPORT DA PROPASALA GÖRE ŞUNU CHANGE BUNU EKLEDİK VS GİBİ ŞEYLER YAZILACAK EĞER DEĞİŞTEYSE.

HAFTAYA SINIFTA TEKRAR KONUŞULACAK YAZDIĞIMIZ ŞEY